MRO Reverse Engineering Study

Assignment #5

Team 13

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Contents

1 Thermal Control Subsystem

The Thermal Control Subsystem (TCS) controls the heat exchange in Mars Reconnaissance Orbiter (MRO) to maintain the temperature of each component within their acceptable temperature ranges. Given its 3pm Local Solar Time (LST) Sun-synchronous orbit [7], MRO experiences less thermal variation compared to orbits closer to local noon, however, the spacecraft still experiences high solar exposure throughout the mission's duration. During nominal operations, MRO also encounters eclipses that result in large temperature swings. Furthermore, throughout the aerobraking phase, MRO is subjected to very high mechanical and thermal loads. For these reasons, the TCS of MRO employs a wide suite of techniques for thermal control. Said techniques and the specifics of subsystems employing thermal control are outlined in the following sections.

1.1 Thermal Requirements of Onboard Systems

The spacecraft (S/C) components are characterized by an operational and a survival temperature limit. The most critical values of each subsystem are reported in Table 1:

	Temperature [°C]				
Component	**Min. Surv.**	**Min. Oper.**	**Max. Oper.**	**Max. Surv.**	**Notes / References**
Instruments					
MARCI	-80	-40	70	100	[16]
HiRISE	—	-110	20	—	[17]
CRISM	—	-163	-148	175	[31, 20, 5]
CTX	—	17	25	—	[10]
SHARAD	—	-50	100	—	Estimated
ONC	—	-30	40	—	Estimated* [11, 15]
TTMTC					
Antennas	-120	-100	100	120	Typical [12]
Antenna gimbals	-50	-40	80	90	Typical [12]
AOCS					
RWs	—	-30	70	—	[8]
Sun sensors	—	-45	85	—	Estimated* [1]
Star trackers	—	-30	60	—	[13]
MIMUs	—	-30	65	—	[9]
Propulsion					
Propellant tank and lines	5	15	40	50	Hydrazine, Typical [12]
Power					
Solar arrays	—	-150	175	200	[22]
Batteries	-25	-10	18	25	[19, 30]

Table 1: Operational and survival temperature ranges for MRO components.
*: based on similar equipment. —: unavailable or inapplicable.

The propulsion system's survival and operational temperatures exceed those of the rest of the S/C, necessitating dedicated heaters. Some instruments also operate outside the S/C's standard range. CRISM needs extreme cold for infrared (IR) imaging; CTX functions within a narrow, and comparatively higher temperature range than the S/C as a whole; and HiRISE is susceptible to thermal damage at high temperatures. These conditions are managed using cryocoolers, heaters, and radiators.

1.2 Architecture and rationale

To meet the thermal requirements across components, the system incorporates a combination of active and passive elements based on conduction and radiation. Their roles are briefly described below.
Radiators: used for dissipating heat from the heat-generating components MRO into deep space. Radiators are, when possible, directly attached to said components, absorbing their excess heat via conduction and dissipating it via radiation. Among the types utilized on MRO are aluminum, silver teflon and painted radiators [29].
Surface Coatings: these passive elements are used to modify the effects of radiation on the S/C surfaces, allowing to selectively determine how much heat is absorbed or reflected from, or radiated to the environment [21].
Multi-Layer Insulation (MLI) or Thermal Blankets: typically made of multiple layers of Mylar or Kapton separated by spacers, their main objective is to create an insulating layer around the component or spacecraft. They also incorporate surface coatings depending on thermal requirements.
Heaters: they are formed by wires capable of releasing heat due to their high electrical resistance. They are directly placed on and around units of MRO that require heating during the mission [21].

1.2.1 Payload

The instruments on board require thermal control components to work properly and maintain the operating temperatures listed in Table 1. Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) operates at very cold temperatures to limit background radiation, being an IR instrument. While passive cooling is involved in keeping the housing of CRISM thermally stable via a graphite epoxy radiator [18], the solar-exposure-rich orbit of MRO necessitates some active cooling. This was achieved using three redundant cryocoolers, all of which are since expired [20], leaving CRISM only able to make near-IR observations. Additionally, being the most thermally sensitive instrument, it was shielded by a cover until the end of Aerobraking and Transition Phase (ATP). The SHAllow Subsurface RADar (SHARAD) instrument uses its support structure as a radiator for its thermal control system, which includes temperature sensors and its own heaters in order to keep its electronics in the operating temperature range [27]. The HIgh Resolution Imaging Science Experiment (HiRISE) relies on black Kapton MLI covering the outside of its heat-sensitive M55J graphite epoxy baffle. Context Camera (CTX) is also covered by MLI, while Mars Color Imager (MARCI) relies on an aluminum radiator [5].

1.2.2 Solar Arrays

		Thickness (m)	
Layer	**Material**	**Inboard Panel (T-0190)**	**Outboard Panel (T-0110, T-0309)**
1	Kapton	7.6×10^{-5}	7.6×10^{-5}
2	M55J/RS-3	5.0×10^{-4}	2.5×10^{-4}
3	Al HC 1.6 PCF	2.9×10^{-2}	2.26×10^{-2}
4	M55J/RS-3	5.0×10^{-4}	2.5×10^{-4}
5	Kapton	5.0×10^{-5}	5.0×10^{-5}

Table 2: Material and thickness of solar array layers

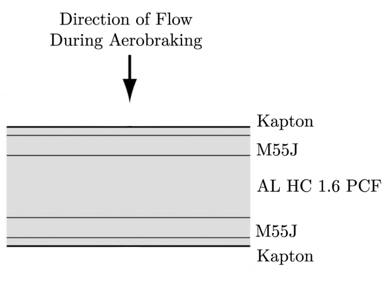

Figure 1: Cross-section [14]

The two solar panels have a total surface area of 20.5 m^2 housing photovoltaic cells, each equipped with a two-axis gimbal system. The solar arrays function as primary drag surfaces during the aerobraking phase of the mission, which results in very high aerodynamic heat loads. The solar array is thus reported as the highest-risk component of MRO during aerobraking [5]. The MRO solar arrays are composed of a multilayered structure, as shown in Figure 1, designed to withstand the thermal stresses, characterized by the materials and thicknesses shown in Table 2.

1.2.3 Batteries

The main power batteries of MRO are two Eagle-Picher 50 Ah, Individual Pressure Vessel (IPV) Ni-H_2 batteries, composed of 11 cells each [28]. The discharge efficiency and supplied voltage of Ni-H_2 batteries are highly temperature-dependent, tapering off with increasing temperature [26]. The charge rates are nearly unaffected by temperature within the operating range [26]. The Ni-H_2 batteries show excellent performance at low temperatures (of about -10 °C), and have been demonstrated to survive in fully frozen spacecraft [30].

1.2.4 High Gain Antenna

The 3-meters High-Gain Antenna (HGA) acts as one of the primary drag surfaces during aerobraking, in addition to its main function of serving the telecommunication goals of the mission. To protect the HGA during aerobraking and orbital temperature swings without hindering communications, it is covered with germanium-coated Kapton [29]. Germanium is preferable in covers for such antennas, as it does not attenuate radio waves. Dedicated heaters keep the Germanium Kapton shielded gimbal mechanism within its operating temperature range [3].

1.2.5 Propulsion Subsystem

The propulsion subsystem of MRO is integrated along the central bus of the spacecraft, with the main thruster assembly extending from aft side. Hydrazine fuel and helium pressurant both exhibit significant sensitivity to temperature variations, prompting the use of a thermally controlled design that leverages both passive insulation layers and heaters to regulate conditions within the propulsion module. Particularly, the propellant lines and tank need to be kept warm to maintain hydrazine above its freezing point and ensure proper thruster operation. The likely solution to meet said requirements (as it is not publicly available) is heaters installed on each tank and line to hold hydrazine above freezing temperature and MLI blankets applied to the tank surfaces to minimize radiative exchange with the surrounding subsystems, as has been the case in some similar missions [24].

1.3 Thermal Fluxes and Phases Correlation

MRO is subjected to various heat sources and sinks throughout its mission. These are summarized in the following table.

	Launch	**Cruise**	**MOI**	**Aerobraking**	**PSP**	**Safe Mode**
Heat Sources						
Solar flux	•	•	•	•	•	•
IR from Earth	•					
IR from Mars			•	•	•	•
Albedo	•*		•	•	•	•
Atmospheric drag				•	•	
Internally generated	•	•	•	•	•	∼
Heat Sinks						
Deep space	•	•	•	•	•	•

Table 3: Presence of heat sources and sinks across different MRO mission phases.
*: From Earth and Moon. ∼: Low to none.

1.4 Reverse Sizing

This section aims to provide a rough order of magnitude thermal analysis of the MRO mission. To do so, the entire S/C is modeled as a single thermal node in steady-state conditions. This simplified time-invariant model does allow the inclusion of a detailed MLI model. It is instead assumed that the entire S/C surface is covered with *"Gold" Kapton*, with known emissivity ε_{SC} and absorptivity α_{SC} values [29].

The assumption of thermal steady-state implies that the total heat entering the S/C equals the amount radiated away. Under this condition, the only unknown in the power balance is the spacecraft temperature, T_{SC}, which is elaborated on in the following sections.

As a starting point, a safety temperature range is derived from Table 1. The safety limits used to define T_{max} and T_{min} are based on the average of the maximum and minimum operational temperatures, excluding the extreme outliers with dedicated thermal controls, in this case, CRISM. This approach helps avoid over-dimensioning the radiator area or the required heating power, which would result from setting overly conservative boundaries. In practice, any individual component or subsystem with operational ranges that are outside of this range are managed using dedicated thermal control systems. To simplify the calculation of cross-sectional areas A_{cross}, the spacecraft is approximated as a thermally equivalent sphere. Subtracting solar panels, antennas, and other external devices, the main body of MRO has a surface area of $A_{SC} = 24.486\,\mathrm{m}^2$, from which $r_{\mathbf{sphere}}$ is computed.

T_{min} w/ margins [°C]	T_{max} w/ margins [°C]	$r_{sphere}[\mathrm{m}^2]$	ε_{SC}	α_{SC}	ε_{rad}
-33 (-48 + 15)	62 (77 - 15)	1.396	0.73	0.41	0.85

Table 4: Sizing Parameters

The thermal inputs considered in this analysis are: solar radiation, albedo power, infrared radiation emitted by Mars, aerodynamic drag heating, and internal power dissipation from active subsystems. The heat generated by air drag power comes from the friction between the external surfaces of the S/C and the gas of the atmosphere around it, which in turn is dependent upon the velocity of S/C and the density of the atmosphere, as can be inferred from the corresponding equation below [4].

$$\text{Atmospheric Drag Heating:} \qquad q_{drag} = \frac{1}{2} \cdot \rho_{atm} \cdot V^3 \cdot C_{H,\text{ incident}}, \qquad Q_{drag} = A_{cross}^{drag}\, q_{drag}$$

The coefficient of convective heat transfer due to aerodynamic friction $C_{H,\text{ incident}} = 0.835\,\mathrm{m}^2$ is taken from the paper by J. H. Prince et al. [25], wherein the real heat flux during the aerobraking maneuver of MRO is measured. The density of Mars atmosphere as a function of the S/C altitude is then calculated using a Mars atmosphere model [6]. The A_{cross}^{drag} is considered to be equal to the A_{cross}^{Sun} because both phenomena can be seen as having an effective area equal to the circle with a radius equal to the one of the equivalent sphere.

The power due to internal heat generated by the spacecraft subsystems is estimated from their respective electrical power consumptions, evaluating independently their electrical-to-thermal efficiencies.

Every object with a temperature above 0 K emits radiation, mostly concentrated in the IR spectrum. The MRO is no exception, as it constantly radiates heat into the colder deep space ($T_{space} = 3\,\mathrm{K}$). The associated heat flux is directed outward, opposite the thermal influx previously described. It can be derived from the Stefan-Boltzmann equation.

The choices of hot and cold cases are described in the next section, where the heating powers inside Table 5 will be analyzed properly.

Phase	$Q_{\textbf{Sun}}$	$Q_{\textbf{ir}}$	$Q_{\textbf{albedo}}$	$Q_{\textbf{drag}}$	$Q_{\textbf{int}}$	$Q_{\textbf{tot}}$
Hot Case	1485.532	514.387	417.940	9193.200	280.930	1.189e+04
Cold Case	-	384.9	-	0.02	93.9	478.9

Table 5: Heating Powers of Hot and Cold Cases (in Watts)

1.4.1 Hot Case

Across the MRO mission, three phases were considered as potential candidates for the occurrence of peak total power due to heat affecting the S/C: Launch and Early Orbit Phase (LEOP), ATP, and Primary Science Phase (PSP). In the LEOP phase, the S/C is at its closest point to the Sun, therefore Q_{Sun} reaches its maximum value during this phase. However, both Q_{int} and Q_{drag} remain low. Nevertheless, a breakdown of LEOP as the hot case is outside scope of this analysis, as the S/C is still attached to the launcher in parts of this phase. As such, a far more complex thermal analysis would be required. During the PSP, all instruments on board are operational. The spacecraft's proximity to Mars leads to an increase in both Q_{albedo} and Q_{IR}. Since the objective is to identify the worst-case (i.e. peak temperature) scenario, Mars was assumed to be at its perihelion and the S/C at its pericenter around Mars respectively. In the case of the aerobraking phase, the Mars's average distance from Sun was used (due to the lengthy duration of this phase), at the pericenter of the spacecraft's orbit. Despite most instruments being turned off, the high value of Q_{drag} due to the very low altitude (as low as 111 km) and the high relative velocity, this phase the one in which the highest thermal loads are expected. As such, it is determined as the hot case for this analysis, as it can be seen in Table 5. The results for the hot case scenario are as follows:

$T_{\text{max, SC}}$ [^{c}ircC]	$A_{\text{rad, body}}$ [m^2]	$A_{\text{rad, deployable}}$ [m^2]
96.9	45.2	6.4

Table 6: Hot Case Results

1.4.2 Cold Case

Several alternatives were considered for the cold case scenario. The selected condition corresponds to the simultaneous occurrence of a solar conjunction and an eclipse. This configuration occurs multiple times during the mission. During this event, the spacecraft is occluded from the Sun by Mars, and from Earth by the Sun, in addition to being at its maximum distance from the Sun. The eclipse results in the absence of solar radiation and albedo in the thermal power balance. On the other hand, the solar conjunction leads to a communication blackout with Earth, which nullifies the power generation associated with the antennas. As for the spacecraft's internally dissipated power, it was considered to be composed of the standby power consumption of the instruments, since most of them cannot operate during an eclipse. Similarly for the sensors, batteries and core electronics. Another important assumption made, consistent with the previous case, is the use of a steady-state solution. This assumption is particularly relevant in the cold case. During the PSP, the eclipse on Mars lasts about 40 minutes. This short duration implies that the spacecraft does not have enough time to cool down to the equilibrium temperature calculated in this analysis. An additional aspect to consider is the behavior of the deployable radiators, which were proposed as a solution in the previous hot case analysis. These radiators are sized for the hot case, but would still passively radiate heat even when it is not desirable to do so, such as during a cold case scenario. This led to the decision to neglect their contribution during phases where eclipse and solar conjunction occur simultaneously. Although this represents a strong assumption, it can still be considered realistic, given the possibility of implementing such a strategy in practice. Several technical solutions would make this feasible, including the use of thermal louvers or deployable radiators designed to open and close multiple times throughout the mission. In the latter case, the steady-state assumption becomes particularly important once again. The consideration of transients however would significantly complicate the thermal model. The main results for the cold case scenario are as follows:

$T_{min,SC}$ [°C]	$Q_{heaters}$ w/ margins [W]
-111.5	2321

Table 7: Cold Case Results

1.4.3 Results Analysis

Table 6 and Table 7 show the main results of the mono-nodal thermal analysis of the MRO's TCS. These results cannot be considered representative of the actual system behavior, furthermore, they exceed the operational limit set at the beginning of the analysis.
In the hot case, the equilibrium temperature reached without the use of radiators is clearly very high. However, this is somewhat expected, as it depends solely on the thermal properties of the spacecraft's external coating and covers, which may not be sufficient in such a harsh environment. The issue becomes apparent when calculating the required radiator surfaces. The estimated area for body-mounted radiators exceeds the usable surface area available on the spacecraft. This discrepancy may be due to an incorrect estimation of the equivalent sphere's surface, derived from the real spacecraft dimensions. In particular, the contribution of the HGA and solar panels, which represent a sizable portion of the spacecraft's total surface, was not included. When deployable radiators are considered instead, the required surface area is still excessive, rendering this solution unfeasible as well. A way to improve these results is increasing radiator emissivity or adjusting the equivalent sphere's coating, either by raising its emissivity or lowering its absorptivity.
Similarly, the cold case results in an equilibrium temperature that is excessively low, below the minimum allowable limit. To mitigate this, heaters were introduced. However, Table 7 shows that the required heater power (already including margin) is too high, making the solution unfeasible. A potential improvement here would be to reduce the spacecraft's emissivity, though this would contradict the recommendations made for the hot case. Another possible refinement would be to model the MLI layers around the spacecraft more accurately, which would improve its thermal insulation.
In both cases, the mono-nodal analysis proves to be insufficient and unreliable, especially considering that the allowed temperature ranges were already defined to be quite broad. A multi-nodal thermal analysis, focusing on the solar panels and HGA (critical in the hot case) or on the scientific instruments (important in the cold case), would likely yield more accurate and realistic results.

1.4.4 Mass and Power Budget

A more detailed TCS modeling is necessary to derive realistic estimates for mass and power budgets. The reverse sizing process has thus far involved only a preliminary system-level analysis, which does not provide sufficient detail to accurately assess mass and power requirements. Based on the values obtained from the reverse sizing and various reference data [23, 2], such as power density and component density, the estimated mass and power budgets (with a margin of 20%), are as follows:

Mass	45.98 kg
Power	2321 W

Table 8: Mass and Power Budget

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Acronyms

AOCS Attitude and Orbit Control Subsystem. 1

ATP Aerobraking and Transition Phase. 2, 5

CRISM Compact Reconnaissance Imaging Spectrometer for Mars. 1, 2, 4

CTX Context Camera. 1, 2

HGA High-Gain Antenna. 3, 6

HiRISE HIgh Resolution Imaging Science Experiment. 1, 2

IR infrared. 1, 2

LEOP Launch and Early Orbit Phase. 5

LST Local Solar Time. 1

MARCI Mars Color Imager. 1, 2

MIMU Miniature Inertial Measurement Unit. 1

MLI Multi-Layer Insulation. 2–4, 6

MRO Mars Reconnaissance Orbiter. 1–6

ONC Optical Navigation Camera. 1

PSP Primary Science Phase. 5

RW reaction wheels. 1

S/C spacecraft. 1, 2, 4, 5

SHARAD SHAllow Subsurface RADar. 1, 2

TCS Thermal Control Subsystem. 1, 6

TTMTC Tracking, Telemetry and Telecommand Subsystem. 1